Class 5: Database Viz with ggplot

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Background

There are many *graphic systems* in R for making plots and figures. These include **"base R graphics"** like the `plot()` and add-on packages like **ggplot2**.

Let's compare how we make a simple graph with these two systems:

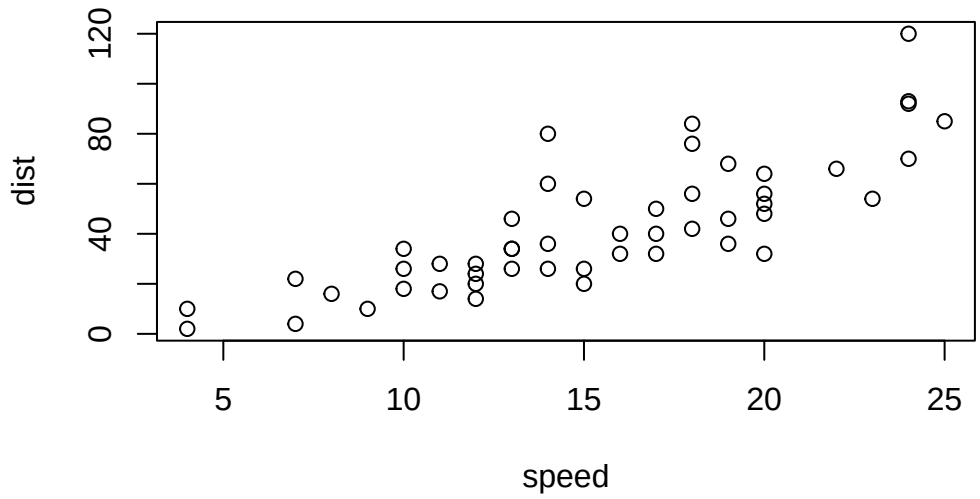
We can use the in-built `cars` dataset:

```
head(cars)
```

```
  speed dist
1     4    2
2     4   10
3     7    4
4     7   22
5     8   16
6     9   10
```

A base R plot:

```
plot(cars)
```



Now for **ggplot2**. Before I can use ggplot2 I need to install it on my computer. To do this we can use the function `install.packages("ggplot2")`

> **N.B** We never run an `install.packages()` in our quarto doc (we can run it in our R console) as it would re-install the package every time I render my document.

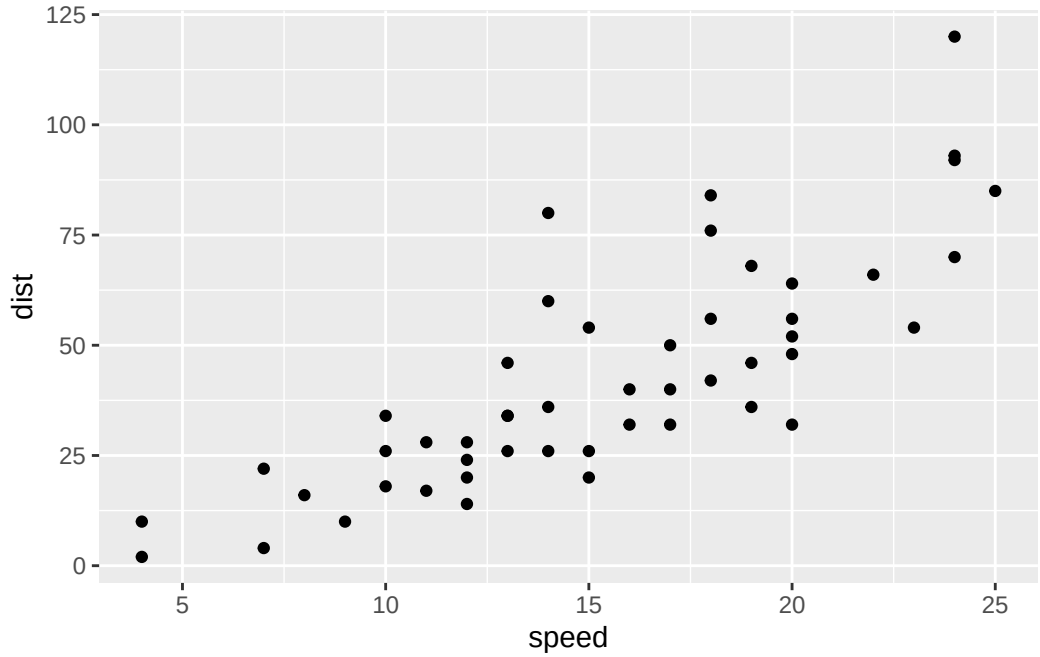
Once installed we still need to load it up the package into our R session. We do this with a `library()` call

```
library(ggplot2)
ggplot(cars)
```

Every ggplot needs at least 3 inputs:

- the **data** (a data.frame of the stuff we want to plot)
- the **aesthetics** (how the data maps to the plot),
- the **geom** layer (how you want the plot drawn, points, lines, etc.)

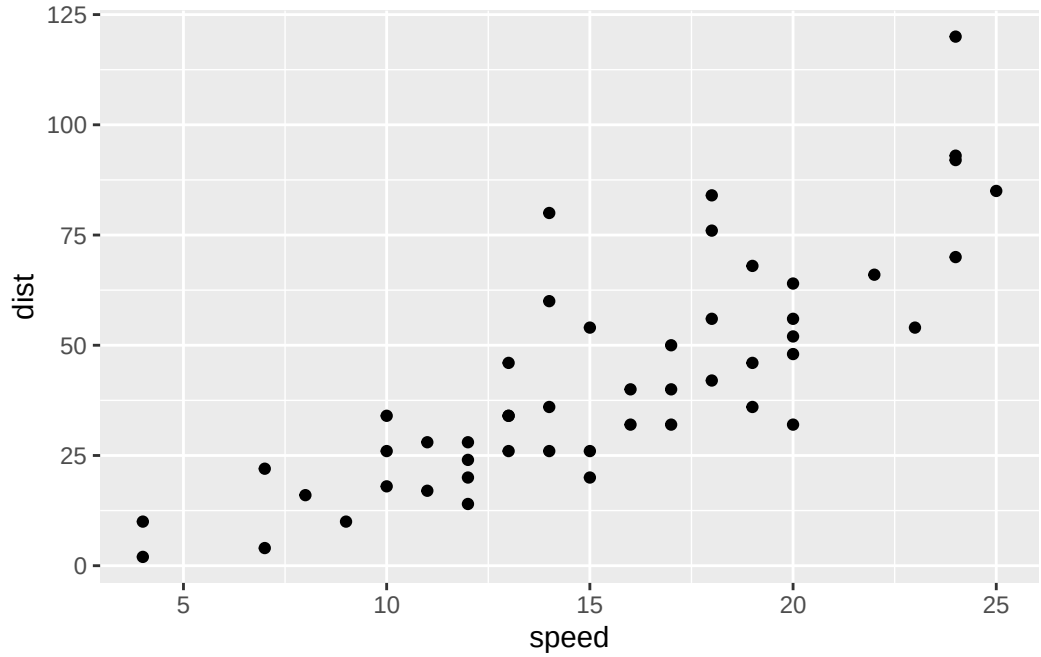
```
ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point()
```



Let's add some features (more layers)

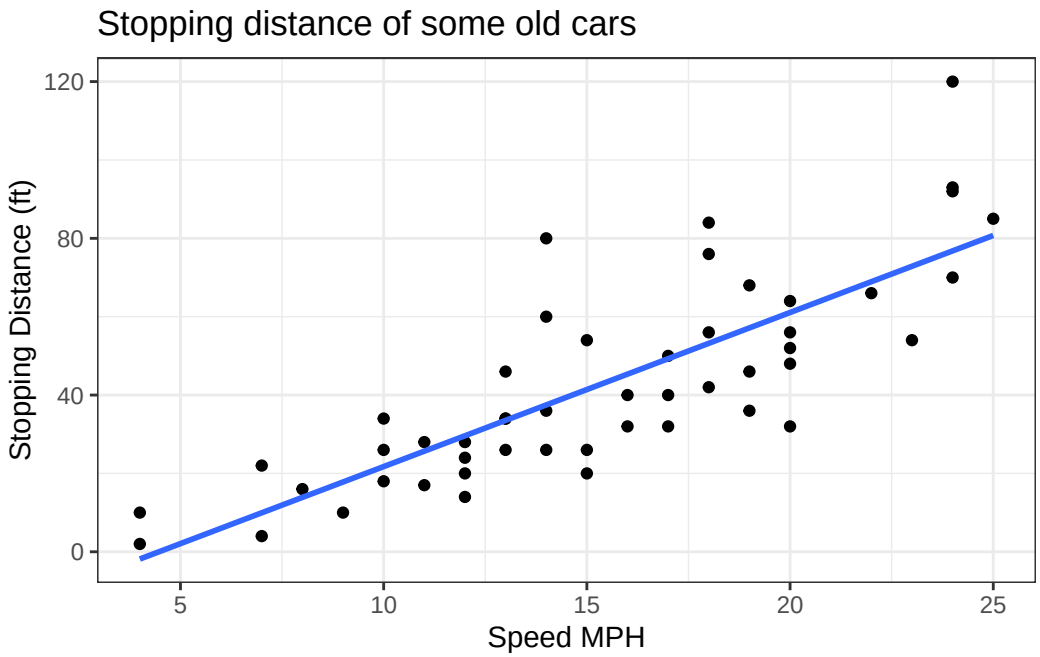
Let's add a trend line that shows the relationship between `speed` and `distance`

```
ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point()
```




```
ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point() +
  geom_smooth(method="lm", se=FALSE) +
  theme_bw() +
  labs(x="Speed MPH",
       y="Stopping Distance (ft)" ,
       title="Stopping distance of some old cars")
```

```
`geom_smooth()` using formula = 'y ~ x'
```



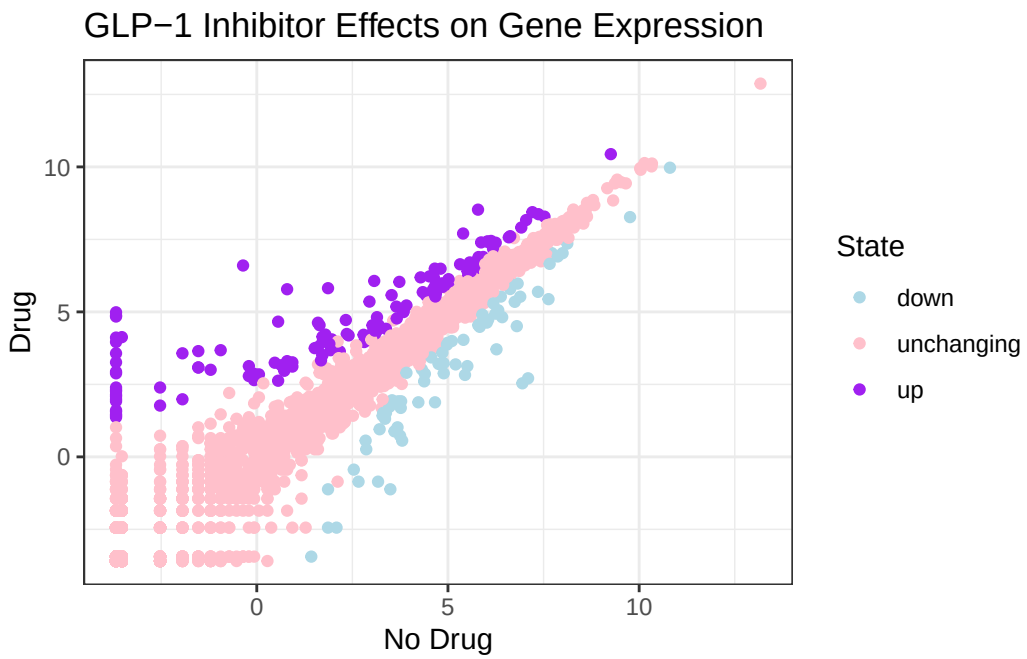
Gene Expression Plot

First we will read some expression data for the effects of a GLP-1 drug

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

```
        Gene Condition1 Condition2      State
1      A4GNT -3.6808610 -3.4401355 unchanging
2       AAAS  4.5479580  4.3864126 unchanging
3      AASDH  3.7190695  3.4787276 unchanging
4       AATF  5.0784720  5.0151916 unchanging
5       AATK  0.4711421  0.5598642 unchanging
6 AB015752.4 -3.6808610 -3.5921390 unchanging
```

```
ggplot(genes) +
  aes(x=Condition1, y=Condition2, col=State) +
  geom_point()  +
  scale_colour_manual( values=c("lightblue","pink","purple") ) +
  theme_bw() +
  labs(x="No Drug", y="Drug", title="GLP-1 Inhibitor Effects on Gene Expression")
```



Q. Use the `nrow()` function to find out how many genes are in this dataset. What is your answer?

```
nrow(genes)
```

```
[1] 5196
```

Q. Use the `table()` function on the `State` column of this data.frame to find out how many 'up' regulated genes there are. What is your answer?

```
table(genes$State)
```

```
      down unchanging         up
        72       4997        127
```

Q. Use the `colnames()` function and the `ncol()` function on the genes data frame to find out what the column names are (we will need these later) and how many columns there are. How many columns did you find?

```
colnames(genes)
```

```
[1] "Gene"       "Condition1" "Condition2" "State"
```

```
ncol(genes)
```

```
[1] 4
```

```
round( table(genes$State)/nrow(genes) * 100, 2 )
```

```
      down unchanging         up
      1.39      96.17       2.44
```

Going Further

Here we read the famous *gapmider* dataset:

```
# File location online
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv

gapminder <- read.delim(url)
head(gapminder)
```

```
      country continent year lifeExp      pop gdpPercap
1 Afghanistan      Asia 1952  28.801  8425333  779.4453
2 Afghanistan      Asia 1957  30.332  9240934  820.8530
3 Afghanistan      Asia 1962  31.997 10267083  853.1007
4 Afghanistan      Asia 1967  34.020 11537966  836.1971
5 Afghanistan      Asia 1972  36.088 13079460  739.9811
6 Afghanistan      Asia 1977  38.438 14880372  786.1134
```

Q. How many entries in this dataset?

```
nrow(gapminder)
```

```
[1] 1704
```

Q. How many different "country" values are in this dataset?

```
length( table(gapminder$country) )
```

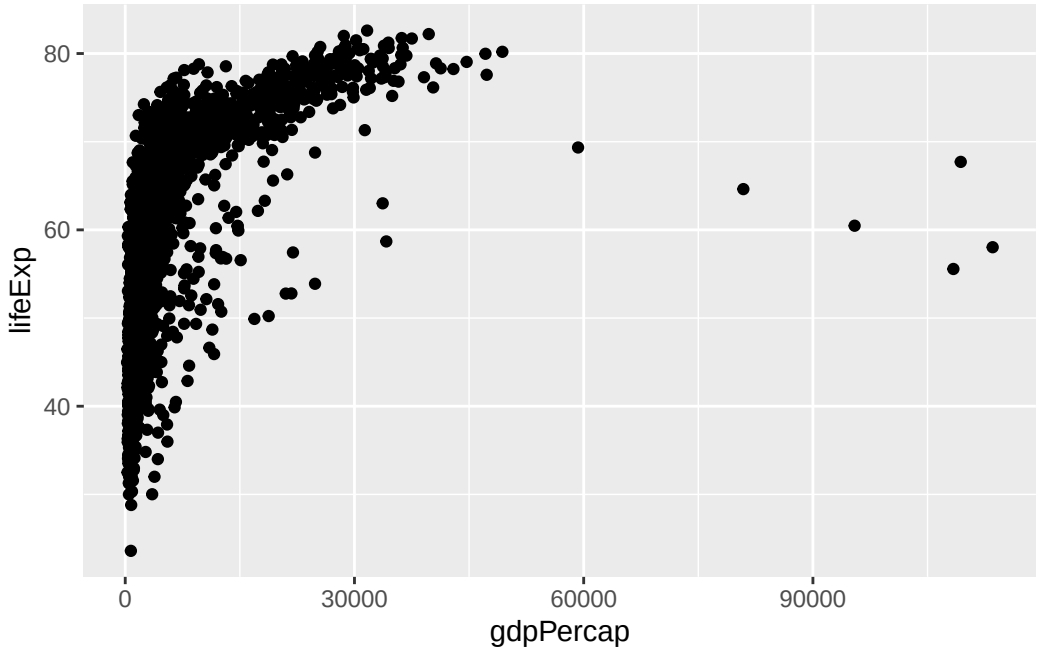
```
[1] 142
```

```
length( unique(gapminder$country) )
```

```
[1] 142
```

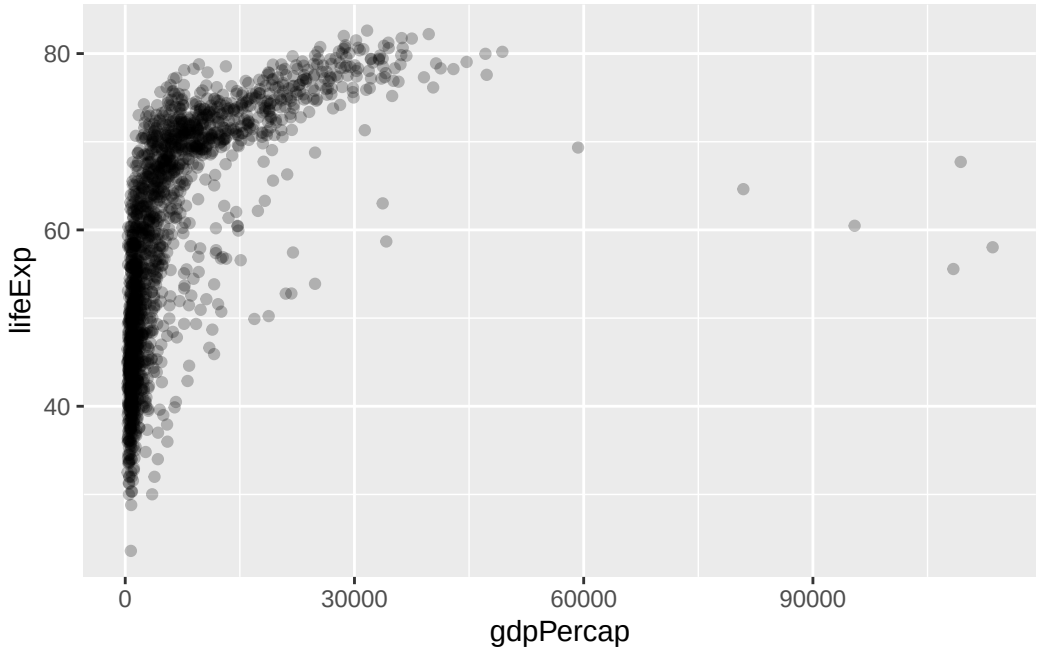
Let's make our first plot of `gdpPercap` vs `lifeExp` :

```
ggplot(gapminder) +
  aes(x=gdpPercap, y=lifeExp) +
  geom_point()
```



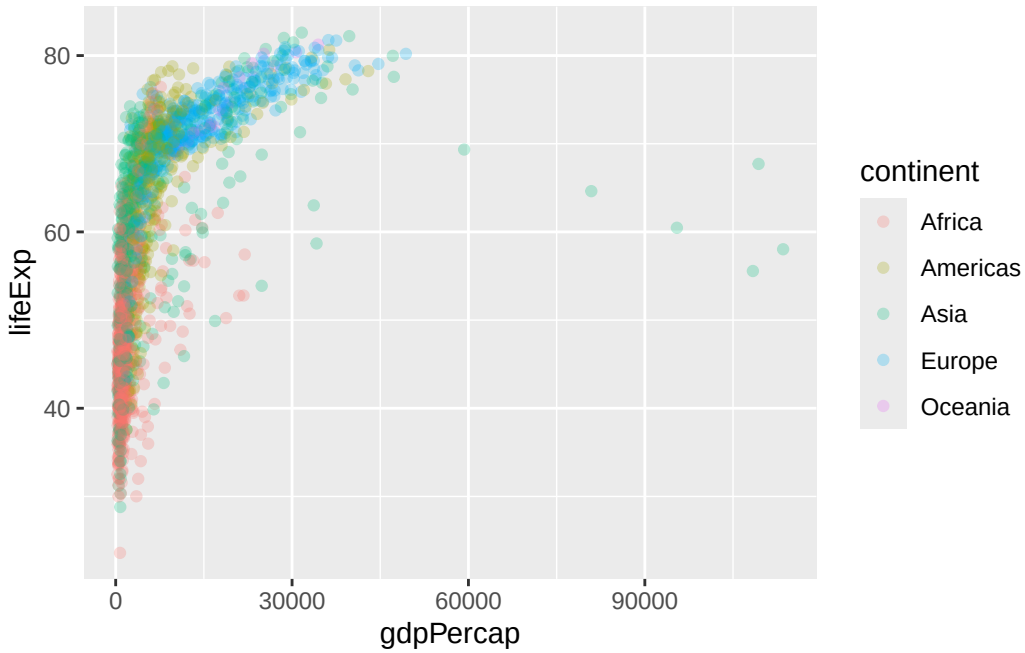
Adding transparency to the points.

```
ggplot(gapminder) +
  aes(x=gdpPercap, y=lifeExp) +
  geom_point(alpha=0.25)
```



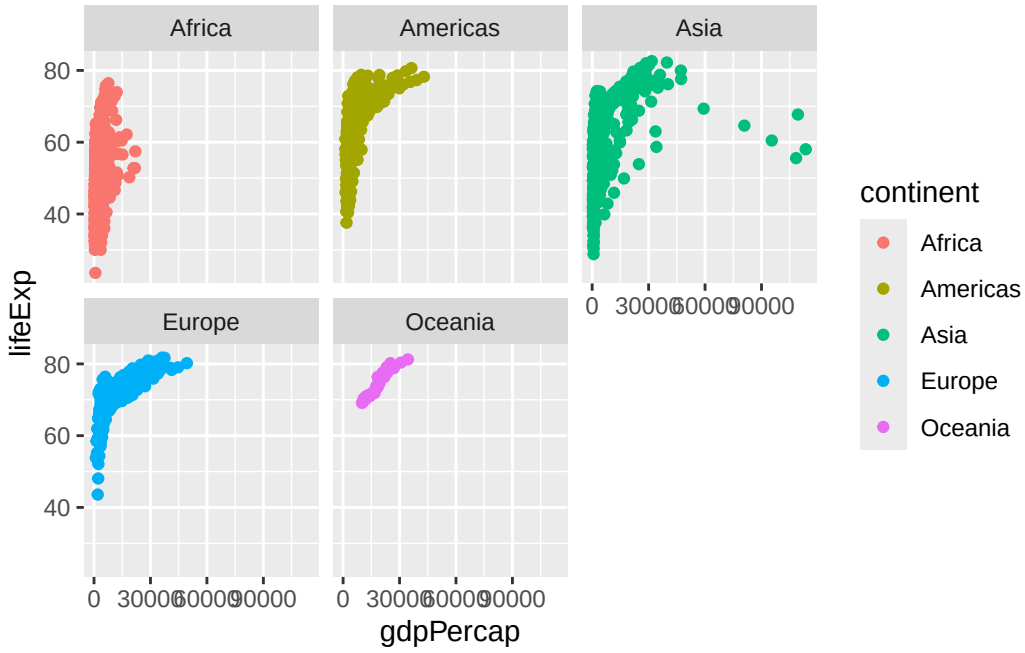
Adding color to the different continents.

```
ggplot(gapminder) +
  aes(x=gdpPercap, y=lifeExp, color=continent) +
  geom_point(alpha=0.25)
```



One useful layer we can add to explore big datasets like this is called "facetting"

```
ggplot(gapminder) +
  aes(x=gdpPercap, y=lifeExp, col=continent) +
  geom_point() +
facet_wrap(~continent)
```



Limit to 1966 and 2004

We can install and use the **dplyr** package to help us "filter" the full dataset down to a smaller subset that only includes the years we want.

First we need to run `install.packages("dplyr")` and then load it with `library(dplyr)`.

```
library(dplyr)
```

```
Attaching package: 'dplyr'
```

```
The following objects are masked from 'package:stats':

    filter, lag
```

```
The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
filter(gapminder, year==2004, country=="Mexico")
```

```
[1] country   continent year      lifeExp   pop       gdpPercap
<0 rows> (or 0-length row.names)
```

```
filter(gapminder, year==1966, country=="Mexico")
```

```
[1] country   continent year      lifeExp   pop       gdpPercap
<0 rows> (or 0-length row.names)
```

```
gapminder$year
```

```
  [1] 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957
 [15] 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967
 [29] 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977
 [43] 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987
 [57] 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997
 [71] 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007
 [85] 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957
 [99] 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967
[113] 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977
[127] 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987
[141] 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997
[155] 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007
[169] 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957
[183] 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967
[197] 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977
[211] 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987
[225] 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997
[239] 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007
[253] 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957
[267] 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967
[281] 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977
[295] 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987
[309] 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997
[323] 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007
[337] 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957
[351] 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967
[365] 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977
[379] 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987
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[393] 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997
[407] 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007
[421] 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957
[435] 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967
[449] 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977
[463] 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987
[477] 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997
[491] 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007
[505] 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957
[519] 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967
[533] 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977
[547] 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987
[561] 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997
[575] 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007
[589] 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957
[603] 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967
[617] 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977
[631] 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987
[645] 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997
[659] 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007
[673] 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957
[687] 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967
[701] 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977
[715] 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987
[729] 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997
[743] 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007
[757] 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957
[771] 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967
[785] 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977
[799] 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987
[813] 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997
[827] 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007
[841] 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957
[855] 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967
[869] 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977
[883] 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987
[897] 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997
[911] 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007
[925] 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957
[939] 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967
[953] 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977
[967] 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987
[981] 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997
```

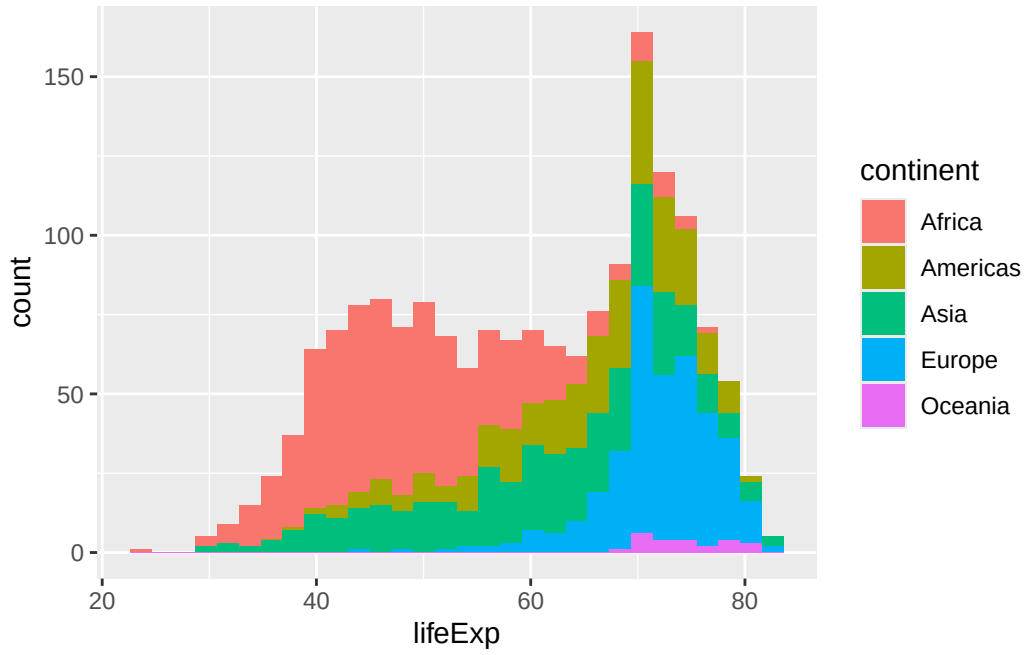
```
 [995] 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007
[1009] 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957
[1023] 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967
[1037] 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977
[1051] 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987
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[1079] 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007
[1093] 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957
[1107] 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967
[1121] 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977
[1135] 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987
[1149] 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997
[1163] 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007
[1177] 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957
[1191] 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967
[1205] 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977
[1219] 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987
[1233] 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997
[1247] 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007
[1261] 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957
[1275] 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967
[1289] 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977
[1303] 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987
[1317] 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997
[1331] 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007
[1345] 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957
[1359] 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967
[1373] 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977
[1387] 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987
[1401] 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997
[1415] 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007
[1429] 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957
[1443] 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967
[1457] 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977
[1471] 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987
[1485] 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997
[1499] 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007
[1513] 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957
[1527] 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967
[1541] 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977
[1555] 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987
[1569] 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997
[1583] 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007
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[1597] 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957
[1611] 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967
[1625] 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977
[1639] 1982 1987 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987
[1653] 1992 1997 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997
[1667] 2002 2007 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007
[1681] 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007 1952 1957
[1695] 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007
```

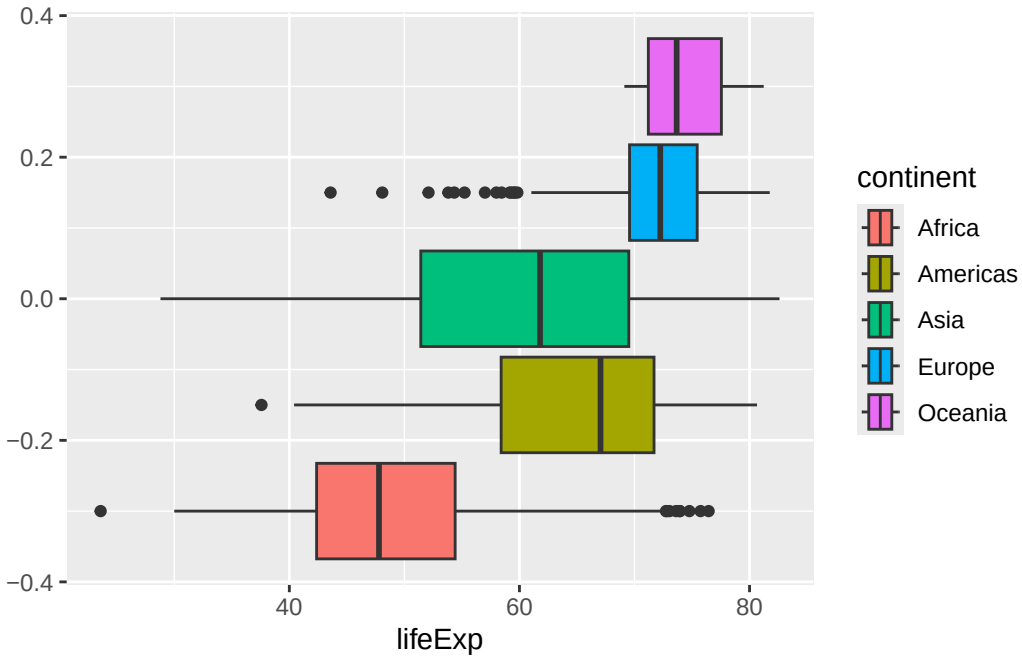
Q. Make a histogram of lifeExp fill color by continent

```
ggplot(gapminder) +
  aes(lifeExp, fill=continent) +
  geom_histogram()
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
ggplot(gapminder) +
  aes(lifeExp, fill=continent) +
  geom_boxplot()
```


0.4
0.2
0.0
−0.2
−0.4
40
60
80
lifeExp
continent
Africa
Americas
Asia
Europe
Oceania